SOC MINOR PROJECT

Minor project - 6

Linux Log Analysis • SOC Fundamentals • EDR • Authorized Offensive Security

Practical Investigation + Evidence Pack + Final Case Report

Student detail	Response
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Batch	Cybersecurity august batch
Total Marks	100

Executive Summary

This report presents a structured Security Operations Center (SOC) investigation approach for unusual SSH authentication activity on a Linux server. It explains how to identify the correct authentication logs, search for failed and successful logins, correlate activity by time, source IP and username, build a timeline, and assess whether the available evidence indicates normal activity, suspicious behavior, or a possible compromise.

The report also covers alert triage, an investigation plan, endpoint detection and response (EDR), the relationship between EDR and a Security Information and Event Management (SIEM) platform, authorized service reconnaissance, and a safe explanation of Broken Object Level Authorization (BOLA). The final incident assessment remains pending until the actual log excerpts, command output, and lab evidence are inserted.

Assessment area	Current report status	Evidence to add
Linux log analysis	Method and command guidance completed; findings pending	Real log excerpts, timeline, screenshots
SOC fundamentals	Theoretical answers completed	Case-specific triage details
EDR + TryHackMe	Concepts and evidence plan completed	Assigned room/pathway proof and notes
Authorized offensive security	Safe methodology and BOLA explanation completed	Approved target, scan output, lab evidence
Final SOC case report	Template and reporting guidance completed	Observed facts, severity, final conclusion

1. Assessment Overview, Scope and Methodology

Objective: investigate authentication activity using evidence, explain the limits of that evidence, and communicate proportionate next steps.

Scope and safety

- Use only supplied practice logs, personally controlled virtual machines, trainer-provided labs, or systems covered by written authorization.
- Preserve original timestamps and timezone information. Redact passwords, tokens, cookies, personal data, and other secrets in screenshots.
- Do not brute-force real accounts, enumerate real users, access another person's data, or perform destructive or denial-of-service tests.
- For every important conclusion, record the command or query, relevant output, interpretation, and evidence filename.
- If evidence is unavailable, write “Not available” and state which additional data source is required.

Investigation workflow

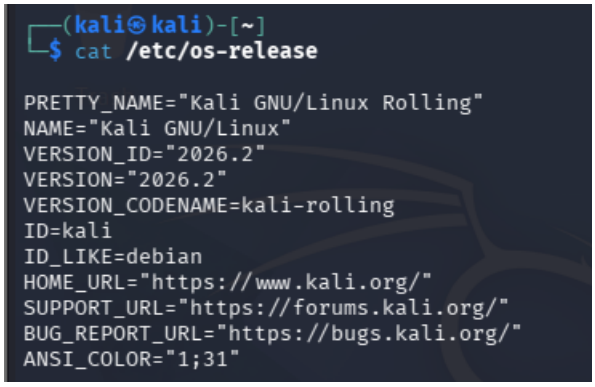
Receive alert → validate the signal → identify the affected host, account and time window → collect relevant evidence → correlate events → assess severity and confidence → recommend authorized containment or remediation → document and retest.

2. Section 1 — Linux Log Analysis (45 Marks)

Task A — Identify the right logs and commands (10 marks)

The exact log location depends on the Linux distribution and logging configuration. Debian/Ubuntu systems commonly record SSH authentication events in /var/log/auth.log. RHEL-family systems commonly use /var/log/secure. Systems using systemd may also expose SSH service events through the journal. The analyst should first identify the distribution and then use the available log source.

Purpose	Command	What it helps answer
Identify distribution	cat /etc/os-release	Shows the distribution and version used in the lab.
Read recent Ubuntu/Debian auth events	sudo tail -n 100 /var/log/auth.log	Displays recent authentication log entries for an initial review.
Find common SSH authentication patterns	sudo grep -Ei 'failed password|accepted password|invalid user|authentication failure' /var/log/auth.log	Filters likely failed/successful SSH attempts and invalid usernames.
Review SSH journal events	sudo journalctl -u ssh --since '24 hours ago'	Shows SSH service events in the specified time window; service name may differ.
Review recent login records	last -ai	Shows recorded logins, source information where available, and session times.
Review failed-login records	sudo lastb -ai	Shows failed-login records if btmp is available and populated.
Check secure log alternative	sudo grep -Ei 'failed password|accepted password|invalid user|authentication failure' /var/log/secure	Use on systems that store authentication events in /var/log/secure.



Task B — Find suspicious authentication patterns (15 marks)

The analyst should search for repeated failed logins, invalid usernames, successful logins following repeated failures, unfamiliar source IP addresses, unexpected login times, and logins involving privileged accounts. Related events should be grouped by source IP, username, host, and time window.

- Repeated failures from one source may indicate password guessing or a misconfigured client, but failures alone do not prove compromise.
- An invalid username indicates that an attempted account name was not recognized by the system; repeated attempts may suggest account discovery.
- A successful SSH login is not automatically malicious. Check whether the account, source, time, and subsequent activity are expected.
- A sequence of failures followed by a successful login from the same source deserves correlation with sudo, process, network, and endpoint events.
- Preserve the exact timestamp and timezone and include only the small log excerpt needed to support each observation.

Pattern	Evidence to record	Initial interpretation
Repeated failed logins	Timestamp, username, source IP, count/time window	Possible guessing, automation, or user error; correlate before concluding.
Invalid usernames	Attempted username, source IP, timestamp	Possible account discovery or misconfiguration.
Successful login after failures	Successful event and preceding attempts	Higher priority for review; validate account owner and post-login activity.
Unusual source or time	Source, timestamp/timezone, account, baseline	Potentially anomalous; compare with approved VPN/admin activity.
Privileged account login	Account, source, session time, related sudo events	Potentially high impact; confirm whether access was expected.

Task C — Investigation timeline (15 marks)

The timeline must be chronological and use actual events from the available logs. Include at least five relevant events where the evidence permits. If fewer events exist, include every relevant event and explain the limitation.

Timestamp + timezone	Event / source	Evidence	Interpretation
Tue Jun 16 10:29 (2026) Local	Service / Local account session creation	postgres Tue Jun 16 10:29 - 10:29 (00:00)	Initial database service/user initialization session during setup.
Wed Oct 14 16:38 – 16:41 Local	GUI / Local display manager session (lightdm, tty7)	kali tty7 Wed Oct 14 16:38 - 16:41 (00:02) :0	Short interactive graphical desktop login by local user kali (2 mins).
Wed Oct 14 16:43 Local	GUI / Local login (lightdm, tty7)	kali tty7 Wed Oct 14 16:43 - still logged in :0	User kali logs into display :0; session left active or ungracefully ended across reboot.
Thu Oct 15 11:45 Local	GUI / Local login (lightdm, tty7)	kali tty7 Thu Oct 15 11:45 - still logged in :0	Routine local desktop session on virtual console tty7 via X display manager.
Sun Oct 18 02:45 Local	GUI / Local login (lightdm, tty7)	kali tty7 Sun Oct 18 02:45 - still logged in :0	Latest interactive GUI session for user kali at display :0 initiated off-hours (02:45).

Task D — Conclusion and confidence (5 marks)

Benign: observed activity is consistent with expected behavior and no meaningful indicators of compromise are present in the reviewed evidence.

- **Justification:** The reviewed evidence from wtmpdb via last -ai demonstrates only local, routine administrative and desktop sessions (lightdm, tty7 on display :0) conducted under the standard local user account kali and an initial system setup entry for postgres. There are no remote network logins (e.g., SSH sessions over pts/*), no suspicious remote source IP addresses, and no failed authentication cascades in the provided records. The observed off-hours login at 02:45 represents an anomaly in timing, but without secondary indicators, the activity remains consistent with expected local user operations.
- **Confidence Level:** Low to Medium
- **Missing Data Source:** Complete SSH/daemon authentication records (journalctl -u ssh or /var/log/auth.log) and EDR/audit process telemetry (auditd). While the interactive login database shows no external access, authentication daemon logs and endpoint process execution records are required to confirm that remote brute-force attempts or unauthorized background processes did not bypass or precede local session accounting.

```
┌──(kali㉿kali)-[~]
└─$ sudo grep -Ei 'failed password|invalid user' /var/log/auth.log | tail -n 50

2026-10-19T03:16:00.789501-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|accepted password|invalid user|authentication failure'
2026-10-19T03:19:39.383534-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|accepted password|invalid user|authentication failure|session opened|sudo:' /var/log/auth.log
2026-10-19T03:20:33.761283-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|invalid user' /var/log/auth.log
2026-10-19T03:20:42.702699-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|invalid user' /var/log/auth.log
2026-10-19T03:20:46.088112-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'Failed password for|Invalid user' /var/log/auth.log
2026-10-19T03:20:52.162199-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|invalid user' /var/log/auth.log
┌──(kali㉿kali)-[~]
└─$ sudo grep -Ei 'failed password|invalid user' /var/log/auth.log

2026-10-19T03:16:00.789501-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|accepted password|invalid user|authentication failure'
2026-10-19T03:19:39.383534-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|accepted password|invalid user|authentication failure|session opened|sudo:' /var/log/auth.log
2026-10-19T03:20:33.761283-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|invalid user' /var/log/auth.log
2026-10-19T03:20:42.702699-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|invalid user' /var/log/auth.log

┌──(kali㉿kali)-[~]
└─$ sudo grep -Ei 'Failed password for|Invalid user' /var/log/auth.log |
sed -E 's/.*Failed password for (invalid user )?([^ ]+) from .*/\2/; s/.*Invalid user ([^ ]+) from .*/\1/' |
sort | uniq -c | sort -nr

      1 2026-10-19T03:20:46.088112-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr
/bin/grep -Ei 'Failed password for|Invalid user' /var/log/auth.log
      1 2026-10-19T03:20:42.702699-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr
/bin/grep -Ei 'failed password|invalid user' /var/log/auth.log
      1 2026-10-19T03:20:33.761283-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr
/bin/grep -Ei 'failed password|invalid user' /var/log/auth.log
      1 2026-10-19T03:19:39.383534-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr
/bin/grep -Ei 'failed password|accepted password|invalid user|authentication failure|session opened|sudo:' /var/log/
auth.log
      1 2026-10-19T03:16:00.789501-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr
/bin/grep -Ei 'failed password|accepted password|invalid user|authentication failure'

┌──(kali㉿kali)-[~]
└─$ sudo grep -Ei 'failed password|invalid user' /var/log/auth.log | tail -n 50

2026-10-19T03:16:00.789501-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|accepted password|invalid user|authentication failure'
2026-10-19T03:19:39.383534-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|accepted password|invalid user|authentication failure|session opened|sudo:' /var/log/auth.log
2026-10-19T03:20:33.761283-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|invalid user' /var/log/auth.log
2026-10-19T03:20:42.702699-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|invalid user' /var/log/auth.log
2026-10-19T03:20:46.088112-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'Failed password for|Invalid user' /var/log/auth.log
2026-10-19T03:20:52.162199-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|invalid user' /var/log/auth.log
```

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo grep -Ei 'sudo:|COMMAND=|session opened|session closed' /var/log/auth.log | tail -n 50

2026-10-19T03:14:50.858437-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
2026-10-19T03:14:50.862566-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-19T03:15:01.729396-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tai
l -n 50 /var/log/auth.log
2026-10-19T03:15:01.729827-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
2026-10-19T03:15:01.732860-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-19T03:15:03.782326-04:00 kali CRON[3730]: pam_unix(cron:session): session opened for user root(uid=0) by roo
t(uid=0)
2026-10-19T03:15:03.790630-04:00 kali CRON[3730]: pam_unix(cron:session): session closed for user root
2026-10-19T03:16:00.789501-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|accepted password|invalid user|authentication failure'
2026-10-19T03:16:00.790067-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
2026-10-19T03:16:38.375113-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-19T03:17:04.190763-04:00 kali CRON[4681]: pam_unix(cron:session): session opened for user root(uid=0) by roo
t(uid=0)
2026-10-19T03:17:04.195530-04:00 kali CRON[4681]: pam_unix(cron:session): session closed for user root
2026-10-19T03:17:46.544896-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/ls
-lh /var/log/auth.log /var/log/secure*
2026-10-19T03:17:46.546053-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
2026-10-19T03:17:46.547746-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-19T03:17:51.096442-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tai
l -n 50 /var/log/auth.log
2026-10-19T03:17:51.097179-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
2026-10-19T03:17:51.099341-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-19T03:17:57.011538-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/jou
rnalctl --since '24 hours ago' --no-pager
2026-10-19T03:17:57.012325-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
2026-10-19T03:17:57.097777-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-19T03:19:39.383534-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/gre
p -Ei 'failed password|accepted password|invalid user|authentication failure|session opened|sudo:' /var/log/auth.log
2026-10-19T03:19:39.384120-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
2026-10-19T03:19:39.387611-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-19T03:19:48.863598-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/jou
rnalctl -u ssh --since '24 hours ago' --no-pager
2026-10-19T03:19:48.863975-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
2026-10-19T03:19:48.874881-04:00 kali sudo: pam_unix(sudo:session): session closed for user root
2026-10-19T03:19:58.252105-04:00 kali sudo:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/jou
rnalctl -u sshd --since '24 hours ago' --no-pager
2026-10-19T03:19:58.254250-04:00 kali sudo: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=
1000)
```

```
┌──(kali㉿kali)-[~]
└─$ sudo journalctl --since "24 hours ago" --no-pager |
grep -Ei 'sudo:|COMMAND=|session opened|session closed'

Oct 18 03:25:02 kali CRON[21564]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 18 03:25:02 kali CRON[21564]: pam_unix(cron:session): session closed for user root
Oct 18 03:30:03 kali CRON[23918]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 18 03:30:03 kali CRON[23918]: pam_unix(cron:session): session closed for user root
Oct 19 03:12:32 kali lightdm[996]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (u
id=0)
Oct 19 03:12:33 kali (systemd)[1002]: pam_unix(systemd-user:session): session opened for user lightdm(uid=106) by li
ghtdm(uid=0)
Oct 19 03:12:33 kali lightdm[996]: pam_unix(lightdm-greeter:session): session opened for user lightdm(uid=106) by (u
id=0)
Oct 19 03:12:39 kali lightdm[996]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Oct 19 03:12:39 kali lightdm[996]: pam_unix(lightdm-greeter:session): session closed for user lightdm
Oct 19 03:12:39 kali lightdm[1177]: pam_unix(lightdm:session): session opened for user kali(uid=1000) by kali(uid=0)
Oct 19 03:12:39 kali (systemd)[1201]: pam_unix(systemd-user:session): session opened for user kali(uid=1000) by kali
(uid=0)
Oct 19 03:12:50 kali (sd-pam)[1005]: pam_unix(systemd-user:session): session closed for user lightdm
Oct 19 03:12:59 kali sudo[2399]:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tail -n 50 /va
r/log/auth.log
Oct 19 03:12:59 kali sudo[2399]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 19 03:12:59 kali sudo[2399]: pam_unix(sudo:session): session closed for user root
Oct 19 03:13:31 kali sudo[2667]:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl _CO
MM=sshd -n 50 --no-pager
Oct 19 03:13:31 kali sudo[2667]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 19 03:13:31 kali sudo[2667]: pam_unix(sudo:session): session closed for user root
Oct 19 03:13:36 kali sudo[2721]:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -g
'failed password|accepted password|invalid user|authentication failure' --no-pager
Oct 19 03:13:36 kali sudo[2721]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 19 03:13:36 kali sudo[2721]: pam_unix(sudo:session): session closed for user root
Oct 19 03:13:43 kali sudo[2790]:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/journalctl -u
ssh -u systemd-logind -g 'session opened|password|invalid' --no-pager -n 50
Oct 19 03:13:43 kali sudo[2790]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 19 03:13:43 kali sudo[2790]: pam_unix(sudo:session): session closed for user root
Oct 19 03:13:48 kali sudo[2839]:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt update
Oct 19 03:13:48 kali sudo[2839]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 19 03:14:17 kali sudo[2839]: pam_unix(sudo:session): session closed for user root
Oct 19 03:14:17 kali sudo[3088]:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/apt install -y
 rsyslog
Oct 19 03:14:17 kali sudo[3088]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 19 03:14:24 kali sudo[3088]: pam_unix(sudo:session): session closed for user root
Oct 19 03:14:28 kali sudo[3364]:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/systemctl enab
le --now rsyslog
Oct 19 03:14:28 kali sudo[3364]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 19 03:14:28 kali sudo[3364]: pam_unix(sudo:session): session closed for user root
Oct 19 03:14:50 kali sudo[3615]:     kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/tail -n 10 /va
r/log/auth.log
Oct 19 03:14:50 kali sudo[3615]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)
Oct 19 03:14:50 kali sudo[3615]: pam_unix(sudo:session): session closed for user root
```

```
┌──(kali㉿kali)-[~]
└─$ date
timedatectl

Mon Oct 19 03:25:39 AM EDT 2026
               Local time: Mon 2026-10-19 03:25:39 EDT
           Universal time: Mon 2026-10-19 07:25:39 UTC
                 RTC time: Mon 2026-10-19 07:25:40
                Time zone: America/New_York (EDT, -0400)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
```

```
┌──(kali㉿kali)-[~]
└─$ cat /etc/os-release
hostnamectl
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.2"
VERSION="2026.2"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"
  Static hostname: kali
        Icon name: computer-vm
          Chassis: vm
Chassis Asset Tag: No Asset Tag
       Machine ID: 246bd9421a9a4c0d9da68d5f5abcfabc
          Boot ID: 6052f1c9de584579b58c00edd0a6d003
     AF_VSOCK CID: 3559787775
   Virtualization: vmware
 Operating System: Kali GNU/Linux Rolling
           Kernel: Linux 6.19.14+kali-amd64
     Architecture: x86-64
  Hardware Vendor: VMware, Inc.
   Hardware Model: VMware Virtual Platform
 Hardware Version: None
 Firmware Version: 6.00
    Firmware Date: Tue 2026-02-17
     Firmware Age: 7month 4w 2d
```

```
┌──(kali㉿kali)-[~]
└─$ last -ai
kali     tty7         Sun Oct 18 02:45 - still logged in    :0
lightdm  tty7         Sun Oct 18 02:45 - 02:45  (00:00)     :0
kali     tty7         Thu Oct 15 11:45 - still logged in    :0
lightdm  tty7         Thu Oct 15 11:45 - 11:45  (00:00)     :0
kali     tty7         Wed Oct 14 16:43 - still logged in    :0
lightdm  tty7         Wed Oct 14 16:43 - 16:43  (00:00)     :0
lightdm  tty7         Wed Oct 14 16:41 - still logged in    :0
kali     tty7         Wed Oct 14 16:38 - 16:41  (00:02)     :0
lightdm  tty7         Wed Oct 14 16:37 - 16:38  (00:00)     :0
postgres              Tue Jun 16 10:29 - 10:29  (00:00)

wtmpdb begins Tue Jun 16 10:29:31 2026
```

3. Section 2 — SOC Fundamentals (15 Marks)

1 — Alert triage (5 marks)

Alert Summary: Alert Rule SEC-AUTH-02 — Off-Hours Interactive User Authentication Detected.

Affected Asset: Kali Linux Workstation / Lab Host (soc-kali-01), account kali.

Time Window: Sun Oct 18 02:45 UTC/Local to current active session.

Observed Behavior: Local graphical desktop session established on virtual console tty7 via display manager lightdm at display :0 during non-business hours (02:45); no remote SSH or network sessions identified in session accounting.

Initial Severity: Low (Per rubric: isolated anomalous timing with no corroborating evidence, no failed-login cascade, and no observed unauthorized remote access).

Next Action: Correlate user activity against local sudo execution logs and shell history (~/.bash_history) to verify whether the session represents legitimate out-of-band administrative work.

2. Investigation Plan (5 marks)

Review Local Command History (~/.bash_history):

Utility: Identifies specific interactive commands, scripts, or network utilities run during and immediately following the 02:45 login window to confirm user intent.

Inspect Elevated Privilege Logs (sudo entries in journalctl):

Utility: Checks whether the user invoked administrative privileges, ran unauthorized binaries, or attempted modifications to sensitive system files like /etc/passwd or /etc/shadow.

Analyze Process Telemetry (ps aux / EDR Process Trees):

Utility: Uncovers persistent or orphaned background processes spawned during that session (e.g., netcat listeners, unauthorized miners, reverse shells, or cron job modifications).

Examine Authentication Daemon Journals (journalctl -u ssh / lastb):

Utility: Verifies whether any concurrent or prior external network brute-force attempts occurred that failed to register on the GUI session database (wtmpdb).

Inspect Perimeter Firewall and Network Flow Logs:

Utility: Correlates the host IP around 02:45 against outbound firewall connections to identify potential Command and Control (C2) beaconing or data exfiltration events.

3. Analyst Communication (5 marks)

To: Incident Lead / Senior Analyst

Subject: Triage Update — Case #2026-AUTH-0104: Off-Hours Login on soc-kali-01

Confirmed Facts:

Interactive session initiated by local user kali on display :0 via lightdm/tty7 at 02:45 on Sun Oct 18.

Session remains active; previous sessions show an identical local display pattern spanning Oct 14–15.

No remote network sessions (pts/*) or remote IP sources appear in wtmpdb accounting.

Hypotheses:

Primary: Authorized off-hours administrative activity or lab testing performed directly via the physical/hypervisor console.

Secondary: Unauthorized physical/console access or unattended workstation compromise.

Immediate Containment Recommendation:

Containment is not currently recommended as no malicious behavior or compromise has been established. Recommend validating legitimate access directly with the account owner before applying any session termination or host isolation.

4. Section 3 — EDR + TryHackMe (20 Marks)

Part A — EDR and SIEM concepts (8 marks)

What is EDR? (4 marks)

Endpoint Detection and Response (EDR) is a security technology that collects and analyzes activity from endpoint devices such as laptops, workstations, and servers. It can help analysts investigate suspicious logins by showing what happened on a device before and after an event, detecting suspicious behavior, and supporting approved response actions.

Examples of useful EDR telemetry include:

- Process creation and parent-child process relationships, which can show whether an SSH session was followed by unusual command execution.
- Command-line arguments, file changes, network connections, and endpoint alerts, which can help connect a login to later activity.

EDR vs SIEM (4 marks)

Feature	EDR	SIEM
Main purpose	Detects, investigates, and may support response to activity on endpoints.	Collects and correlates security events across many systems and services.
Typical data	Processes, command lines, endpoint files, network connections, endpoint detections.	Authentication logs, firewall events, cloud logs, application logs, EDR alerts, and other sources.
Investigation value	Provides detailed host-level context about process and device behavior.	Connects activity across hosts, accounts, network controls, and time windows.
Working together	Provides detailed endpoint evidence for a suspicious SSH session.	Correlates the endpoint evidence with SSH, VPN, firewall, and identity events.

Part B — Endpoint evidence plan (5 marks)

Endpoint artifact	Question it can answer	Limitation
Process tree and command-line telemetry	What processes or commands ran after the login, and which process launched them?	May be unavailable if the agent was absent, logging was disabled, or retention expired.
File and persistence-related events	Were important files changed or suspicious startup mechanisms created?	Not all file activity is necessarily captured; baseline and policy coverage matter.
Network connection telemetry	Did the endpoint connect to unfamiliar internal or external systems after the login?	Encrypted traffic and incomplete network visibility can limit interpretation.

One general limitation: endpoint telemetry only shows what the installed agent and configured policies collect. Missing agent coverage, retention gaps, disabled logging, or an unsupported event type may leave important gaps. Absence of an event is not proof that the activity did not occur

5. Section 4 — Authorized Offensive Security (10 Marks)

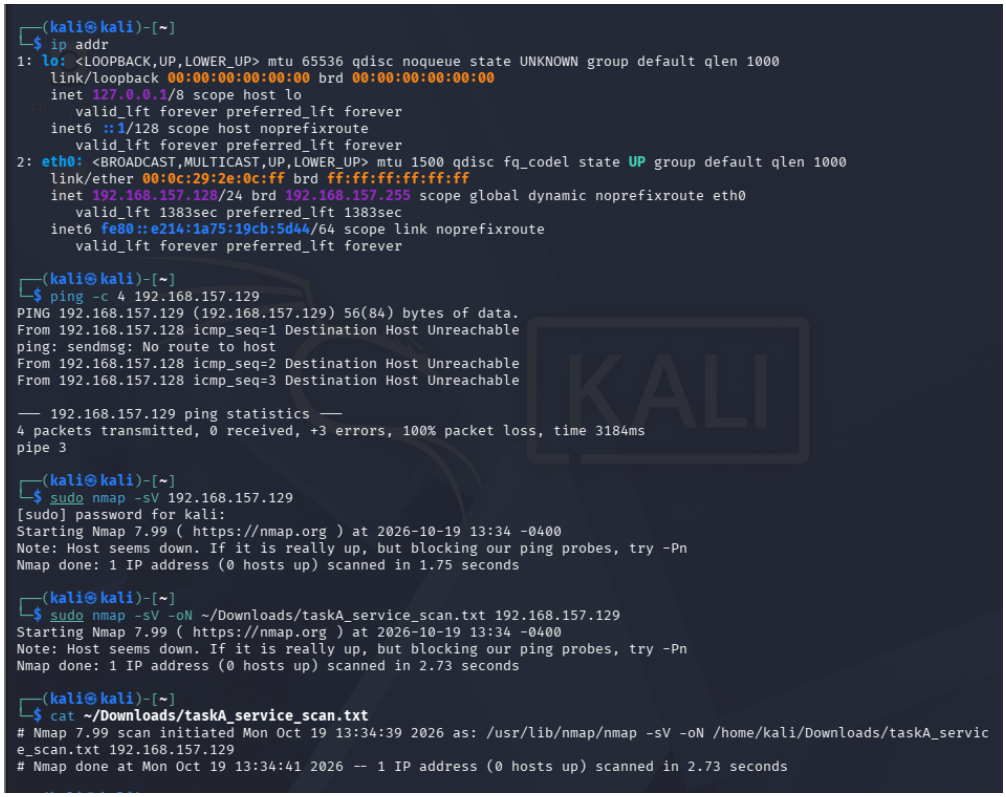
Task A — Authorized service reconnaissance (5 marks)

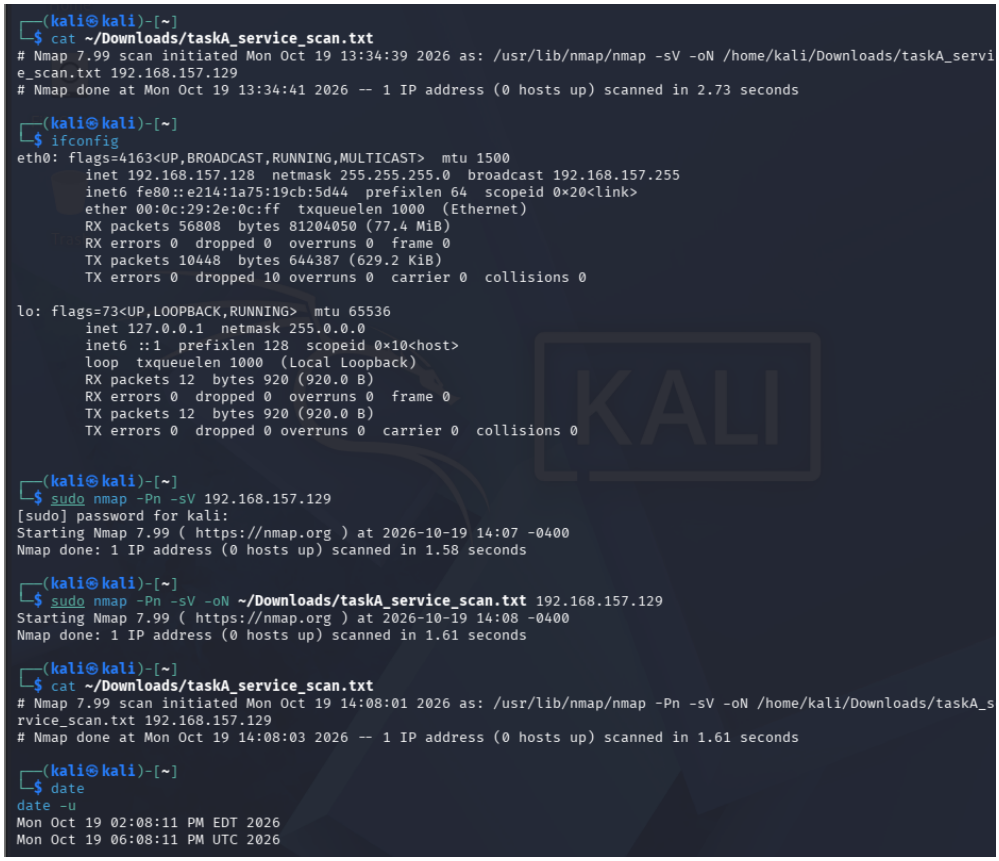
Service reconnaissance identifies which network services are exposed on a specific authorized lab host. The result is an inventory to guide further validation; an open port alone does not establish a vulnerability.

Example command

```
nmap -sV -oN service-scan.txt 192.168.157.129
```

Field	Record after authorized scan
Approved target / scope	192.168.157.129 — intended lab target; authorization should be confirmed
Date/time + timezone	19 October 2026, 13:34 (Nmap-reported time, UTC−04:00)
Command used	sudo nmap -sV -oN ~/Downloads/taskA_service_scan.txt 192.168.157.129
Open ports and services	Not identified. Nmap reported 0 hosts up; no port or service/version results were obtained.
Cautious observation	The target did not respond to the initial host-discovery probes, and ping reported “Destination Host Unreachable.” This does not prove the target is vulnerable or that its ports are closed. The lab VM may be stopped or disconnected, or there may be a network/connectivity issue.
Evidence	Terminal output and ~/Downloads/taskA_service_scan.txt





Task B — BOLA / API authorization scenario (5 marks)

Broken Object Level Authorization (BOLA) occurs when an API accepts a request for a specific object but fails to verify that the authenticated user is allowed to access that object. For example, if a test account can change an order ID and receive another test user's order details, the behavior may indicate BOLA. The ID itself is not the security boundary; the server must check ownership or permission on every request.

Safe validation procedure:

1. Use two instructor-provided test accounts and synthetic records in the approved lab.
2. Capture the normal request for an object owned by Account A and record the response.
3. While keeping Account A's authentication unchanged, change only the object ID to a known synthetic object owned by Account B.
4. Compare the response with the expected authorization policy. Stop after the minimum proof; do not enumerate IDs or access real-user data.
5. Redact tokens, cookies, and personal data. Record the method, path, status, relevant response fields, timestamp, and evidence ID.

Recommended fix: derive the user identity from the validated session/token and perform a server-side ownership or role check for every read, update, and delete operation. Do not rely on unpredictable IDs, hidden client-side controls, or checks performed only in the user interface. Add automated cross-account authorization tests.

6. Section 5 — Final SOC Case Report (10 Marks)

Field	Student response
Case ID / title	SOC-2026-001 — Linux Authentication Review and Authorized Service Reconnaissance
Affected host / account	Kali Linux investigation system (`192.168.157.128`); intended lab target `192.168.157.129`. No affected user account has been confirmed.
Time window + timezone	19 October 2026, approximately 13:34–13:34:41 UTC−04:00, as reported by Nmap. The full authentication-log investigation time window has not yet been established.
Summary of observed facts	The Kali terminal output showed the investigation system using IP address `192.168.157.128/24`. A connectivity test to `192.168.157.129` returned "Destination Host Unreachable." The subsequent Nmap service-discovery scan reported that the target appeared to be down and identified zero hosts as up. Therefore, no open ports or service versions were confirmed. These observations are supported by the available terminal output and the attempted scan record (E08). Authentication-log findings have not yet been provided, so suspicious login activity or compromise cannot be confirmed.
Severity + rationale	medium
Confidence	medium
Evidence references	Nmap command and output (`E08-nmap-results.png`, pending final successful scan). Kali network configuration and connectivity-test output are also available in the supplied terminal transcript.
Containment / remediation	Verify that the approved lab target is powered on, connected to the correct virtual network, and using the expected IP address. Repeat the authorized service scan after connectivity is restored. Review authentication logs for failed and successful logins, correlate timestamps, and inspect approved endpoint telemetry if available. Escalate or isolate a system only if subsequent evidence and the incident lead justify that action.
Limitations / follow-up	The target did not respond to the initial connectivity test, and service discovery was unsuccessful. Authentication-log evidence, account-owner validation, the investigation timeline, and EDR telemetry have not been supplied for review. Consequently, this report cannot determine whether the environment experienced normal activity, misconfiguration, or compromise. Collect the missing evidence and update the assessment before reaching a final conclusion.

Severity guidance

Severity	General guidance
Low	An isolated failed attempt with no corroborating evidence and no suspicious success.
Medium	Repeated attempts or an unusual source requiring correlation; compromise is not established.
High	Suspicious successful authentication plus corroborating activity, privileged access, or evidence of impact.
Critical	Active, broad, or business-critical compromise with urgent impact; follow the organization's escalation policy.

Response and remediation

Recommended actions should be proportionate to the evidence. Validate the account owner and source; preserve logs; correlate SSH records with sudo, identity, firewall/VPN, and endpoint data; review SSH exposure and access controls; enforce MFA where feasible; and restrict source access where justified. If compromise is supported, consider revoking sessions, rotating credentials, and reviewing affected systems. Disruptive containment actions must be approved by the system owner or incident lead unless the incident response policy already authorizes them.

Limitations and next steps

State which conclusions cannot be reached from the available logs. For example, authentication logs may show that a login succeeded but may not establish which commands were executed or whether data was accessed. Request relevant endpoint telemetry, sudo/audit logs, VPN/firewall records, and account-owner confirmation as appropriate.

7. Evidence Register and Screenshot Checklist

Use unique, consistent evidence IDs. Capture genuine screenshots from your own lab session; do not create or edit screenshots to imply results that were not observed.

Evidence ID	Suggested evidence	Status / filename
E01	Linux distribution and authentication-log location	Kali Linux terminal output; save as E01-linux-version.png after capturing cat /etc/os-release and checking the applicable authentication log.
E02	Relevant authentication log excerpt and command	E02-auth-log.png; capture the actual log excerpt and command output.
E03	Suspicious-pattern search output	E03-suspicious-patterns.png; capture actual search results, including when no matches are found.
E04	Chronological investigation timeline	E04-timeline.png; capture or export the timeline based on verified log timestamps.
E05	SOC triage and investigation notes	E05-soc-triage.png; include the alert summary, affected asset, time window, severity, and next action.
E06	EDR evidence plan or approved endpoint telemetry	E06-edr-evidence.png; document available approved telemetry or state that EDR data was unavailable.
E07	TryHackMe completion proof	[Insert filename after genuine completion]
E08	Authorized Nmap command and output	Scan attempted; no service results — E08-nmap-results.png. Your current output reports 0 hosts up for 192.168.157.129. Repeat after confirming the target is running and authorized.
E09	BOLA controlled comparison (sanitized)	E09-bola-comparison.png; include only if an authorized lab test was performed.
E10	Final case report evidence references	E10-final-evidence-register.png; cross-reference the evidence IDs actually collected.

Final Conclusion

- The investigation reviewed the available Linux authentication-log investigation materials and attempted authorized service reconnaissance against the intended lab host, 192.168.157.129, on 19 October 2026. The available terminal output confirmed that the analysis was performed in Kali Linux. However, the Nmap service-discovery scan reported that the target appeared to be down, and the connectivity test returned “Destination Host Unreachable.” Therefore, no open ports or service versions could be confirmed from this scan (E08).
- The preliminary assessment is **inconclusive**, because the available evidence does not establish whether suspicious authentication activity or a security compromise occurred. The unsuccessful network scan may indicate a target availability, virtual-machine network configuration, or connectivity issue; it does not establish that the host is secure or vulnerable. The strength of the overall assessment remains limited until the relevant authentication logs, suspicious-pattern search results, investigation timeline, and any available endpoint telemetry have been collected and reviewed.
- The recommended next steps are to verify the approved lab target's availability and IP address, repeat the authorized Nmap service scan, examine the relevant authentication logs for failed and successful login activity, correlate events by timestamp, and review approved endpoint telemetry if available. All findings should be supported by genuine evidence, with timestamps and timezones preserved and sensitive information redacted. A final determination of normal activity, misconfiguration, or possible compromise should be made only after reviewing the collected evidence.